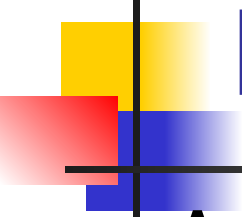
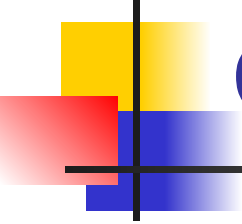


Probability Distributions



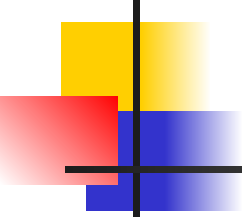
Random Variable

- A random variable x takes on a defined set of values with different probabilities.
 - For example, if you roll a die, the outcome is random (not fixed) and there are 6 possible outcomes, each of which occur with probability one-sixth.
 - For example, if you poll people about their voting preferences, the percentage of the sample that responds “Yes on Proposition 100” is also a random variable (the percentage will be slightly differently every time you poll).
- Roughly, probability is how frequently we expect different outcomes to occur if we repeat the experiment over and over (“frequentist” view)



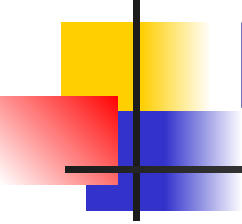
Random variables can be discrete or continuous

- **Discrete** random variables have a countable number of outcomes
 - Examples: Dead/alive, treatment/placebo, dice, counts, etc.
- **Continuous** random variables have an infinite continuum of possible values.
 - Examples: blood pressure, weight, the speed of a car, the real numbers from 1 to 6.

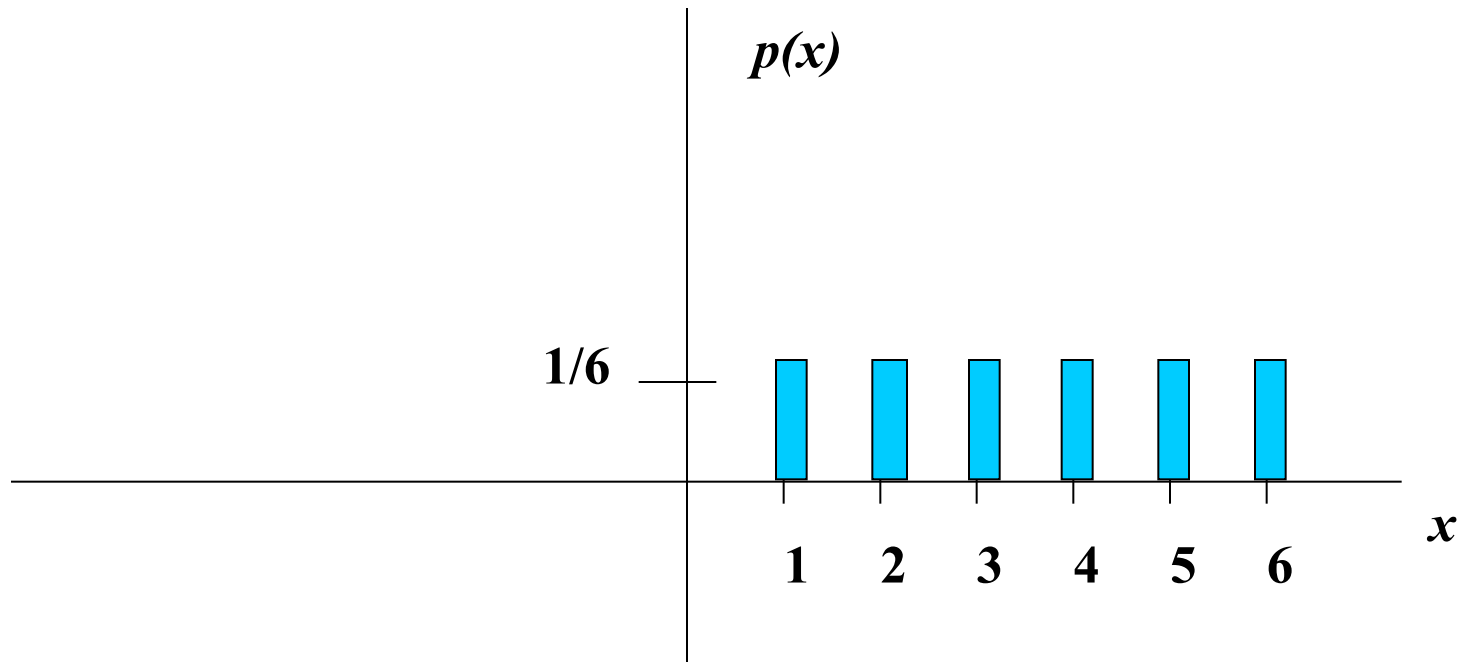


Probability functions

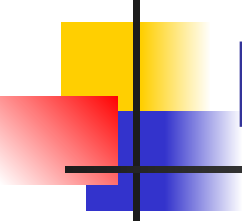
- A probability function maps the possible values of x against their respective probabilities of occurrence, $p(x)$
- $p(x)$ is a number from 0 to 1.0.
- The area under a probability function is always 1.



Discrete example: roll of a die



$$\sum_{\text{all } x} P(x) = 1$$

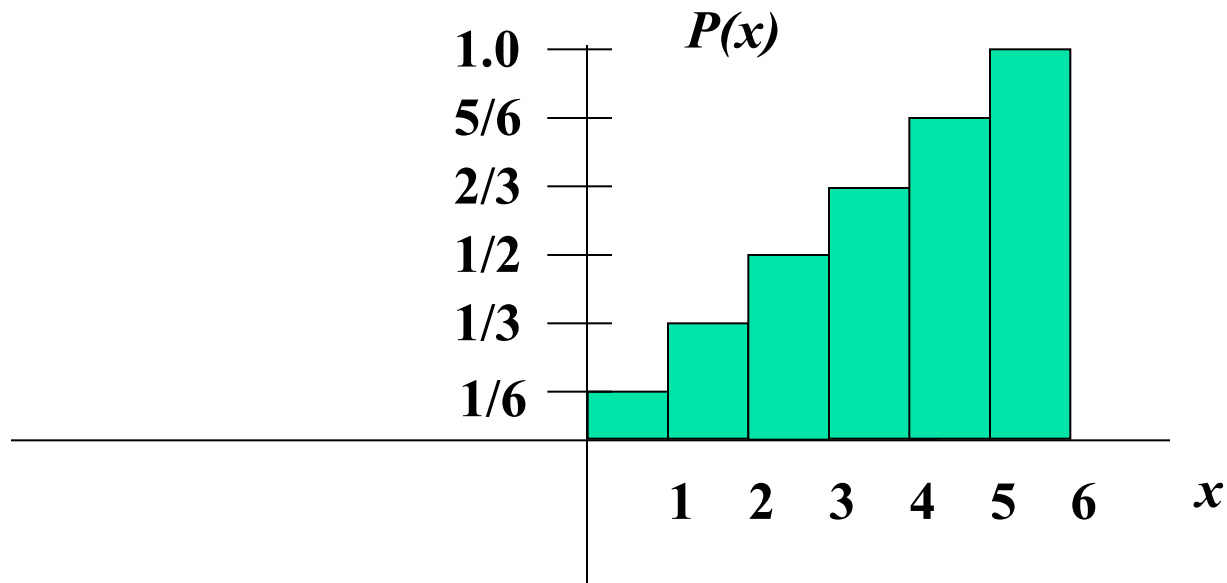


Probability mass function (pmf)

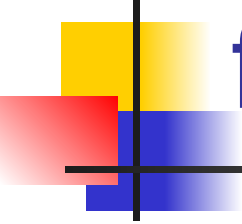
x	$p(x)$
1	$p(x=1)=1/6$
2	$p(x=2)=1/6$
3	$p(x=3)=1/6$
4	$p(x=4)=1/6$
5	$p(x=5)=1/6$
6	<u>$p(x=6)=1/6$</u>

1.0

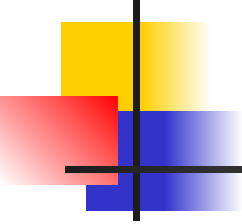
Cumulative distribution function (CDF)



Cumulative distribution function



x	$P(x \leq A)$
1	$P(x \leq 1) = 1/6$
2	$P(x \leq 2) = 2/6$
3	$P(x \leq 3) = 3/6$
4	$P(x \leq 4) = 4/6$
5	$P(x \leq 5) = 5/6$
6	$P(x \leq 6) = 6/6$



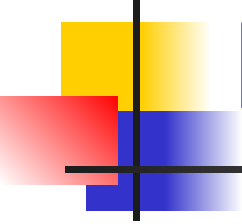
Examples

1. What's the probability that you roll a 3 or less?

$$P(x \leq 3) = 1/2$$

2. What's the probability that you roll a 5 or higher?

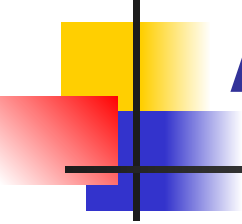
$$P(x \geq 5) = 1 - P(x \leq 4) = 1 - 2/3 = 1/3$$



Practice Problem

Which of the following are probability functions?

- a. $f(x) = .25$ for $x = 9, 10, 11, 12$
- b. $f(x) = (3-x)/2$ for $x = 1, 2, 3, 4$
- c. $f(x) = (x^2 + x + 1)/25$ for $x = 0, 1, 2, 3$



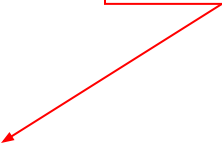
Answer (a)

a. $f(x) = .25$ for $x = 9, 10, 11, 12$

x	$f(x)$
9	.25
10	.25
11	.25
12	<u>.25</u>

1.0

**Yes, probability
function!**



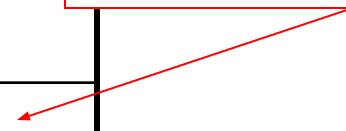


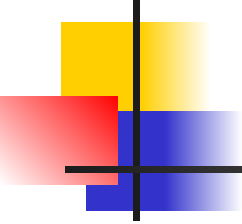
Answer (b)

b. $f(x) = (3-x)/2$ for $x=1,2,3,4$

x	$f(x)$
1	$(3-1)/2=1.0$
2	$(3-2)/2=.5$
3	$(3-3)/2=0$
4	$(3-4)/2=-.5$

Though this sums to 1,
you can't have a negative
probability; therefore, it's
not a probability
function.





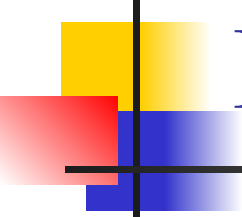
Answer (c)

c. $f(x) = (x^2 + x + 1)/25$ for $x=0,1,2,3$

x	f(x)
0	1/25
1	3/25
2	7/25
3	<u>13/25</u>

24/25

Doesn't sum to 1. Thus,
it's not a probability
function.



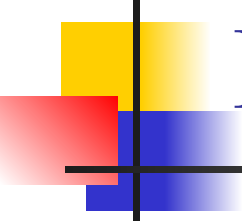
Practice Problem:

- The number of ships to arrive at a harbor on any given day is a random variable represented by x . The probability distribution for x is:

x	10	11	12	13	14
$P(x)$	.4	.2	.2	.1	.1

Find the probability that on a given day:

- exactly 14 ships arrive $p(x=14) = .1$
- At least 12 ships arrive $p(x \geq 12) = (.2 + .1 + .1) = .4$
- At most 11 ships arrive $p(x \leq 11) = (.4 + .2) = .6$



Practice Problem:

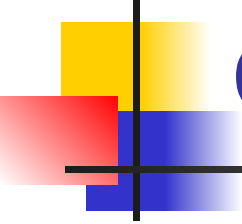
You are lecturing to a group of 1000 students. You ask them to each randomly pick an integer between 1 and 10. Assuming, their picks are truly random:

- What's your best guess for how many students picked the number 9?

Since $p(x=9) = 1/10$, we'd expect about $1/10^{\text{th}}$ of the 1000 students to pick 9. 100 students.

- What percentage of the students would you expect picked a number less than or equal to 6?

Since $p(x \leq 6) = 1/10 + 1/10 + 1/10 + 1/10 + 1/10 + 1/10 = .6$ 60%



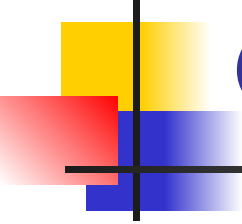
Important discrete distributions in epidemiology...

- Binomial
 - Yes/no outcomes (dead/alive, treated/untreated, smoker/non-smoker, sick/well, etc.)
- Poisson
 - Counts (e.g., how many cases of disease in a given area)



Continuous case

- The probability function that accompanies a continuous random variable is a continuous mathematical function that integrates to 1.
- The probabilities associated with continuous functions are just areas under the curve (integrals!).
- Probabilities are given for a range of values, rather than a particular value (e.g., the probability of getting a math SAT score between 700 and 800 is 2%).



Continuous case

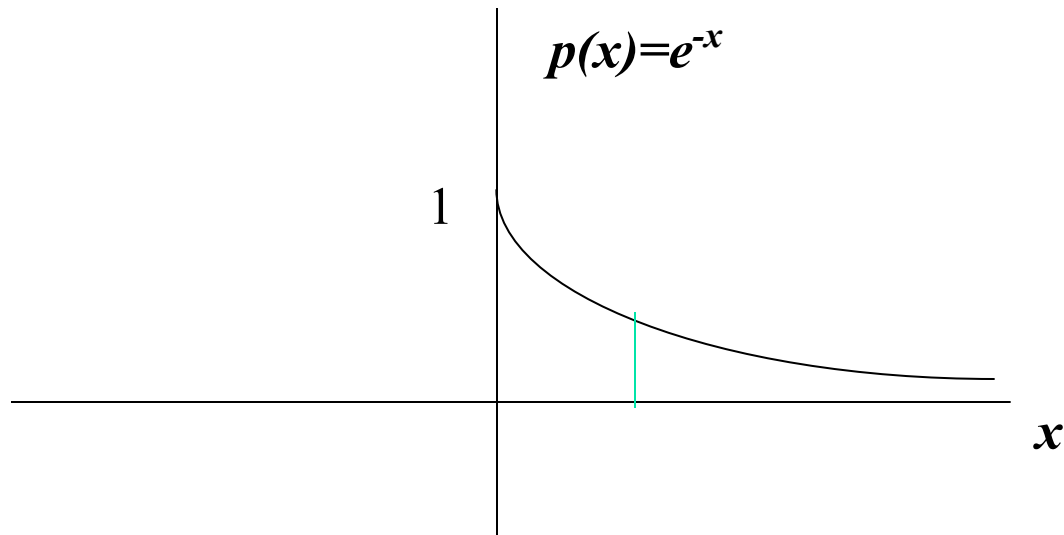
- For example, recall the negative exponential function (in probability, this is called an “exponential distribution”):

$$f(x) = e^{-x}$$

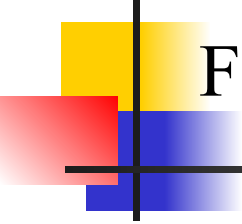
- This function integrates to 1:

$$\int_0^{+\infty} e^{-x} = -e^{-x} \Big|_0^{+\infty} = 0 + 1 = 1$$

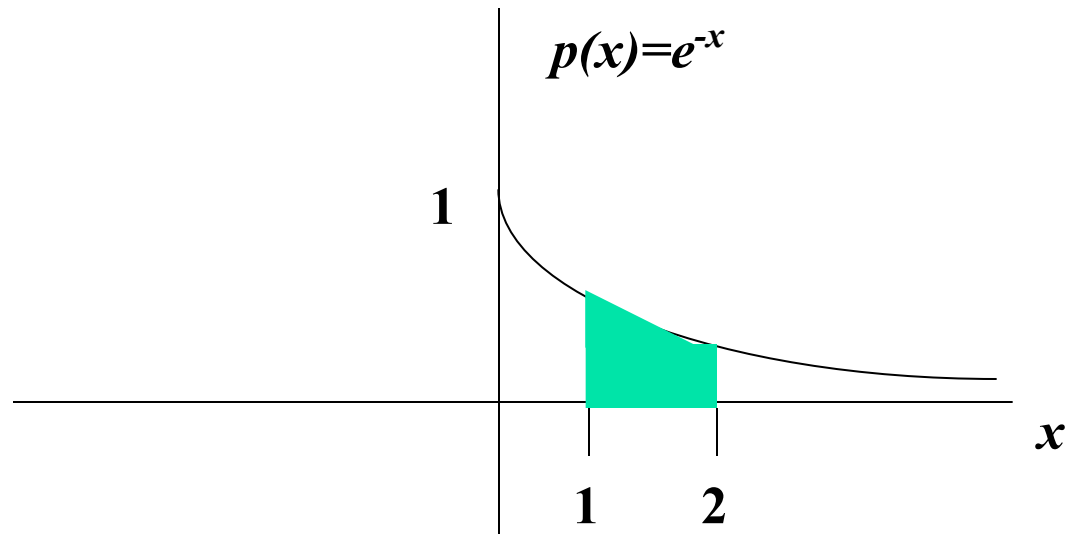
Continuous case: “probability density function” (pdf)



The probability that x is any exact particular value (such as 1.9976) is 0; we can only assign probabilities to possible ranges of x .

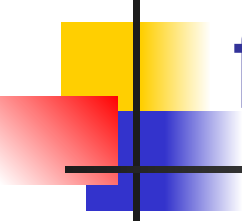


For example, the probability of x falling within 1 to 2:



$$P(1 \leq x \leq 2) = \int_1^2 e^{-x} = -e^{-x} \Big|_1^2 = -e^{-2} - (-e^{-1}) = -.135 + .368 = .23$$

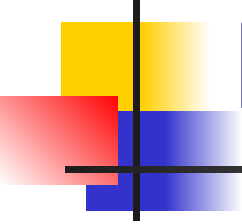
Cumulative distribution function



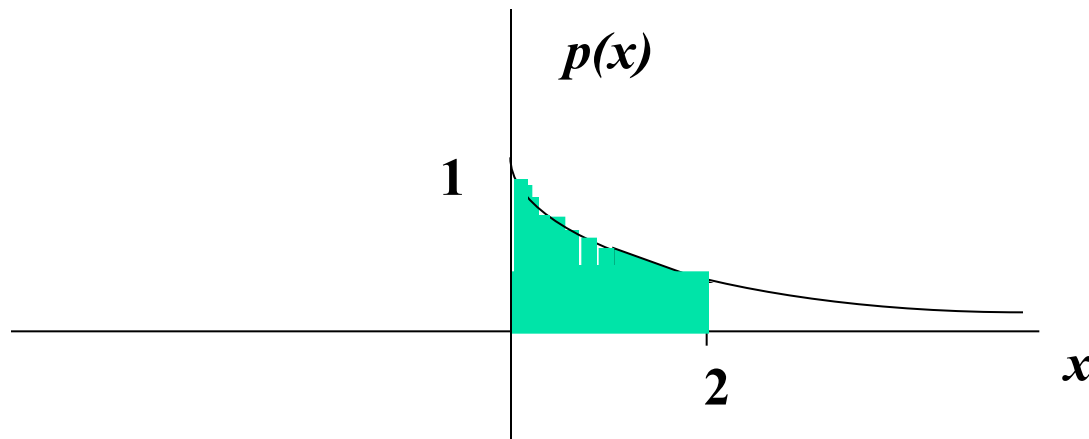
As in the discrete case, we can specify the “cumulative distribution function” (CDF):

The CDF here = $P(x \leq A) =$

$$\int_0^A e^{-x} = -e^{-x} \Big|_0^A = -e^{-A} - (-e^0) = -e^{-A} + 1 = 1 - e^{-A}$$



Example

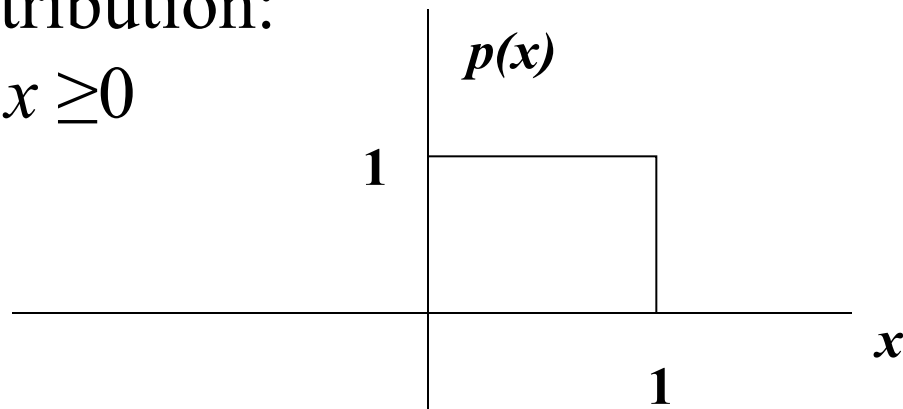


$$P(x \leq 2) = 1 - e^{-2} = 1 - .135 = .865$$

Example 2: Uniform distribution

The uniform distribution: all values are equally likely

The uniform distribution:
 $f(x) = 1$, for $1 \geq x \geq 0$

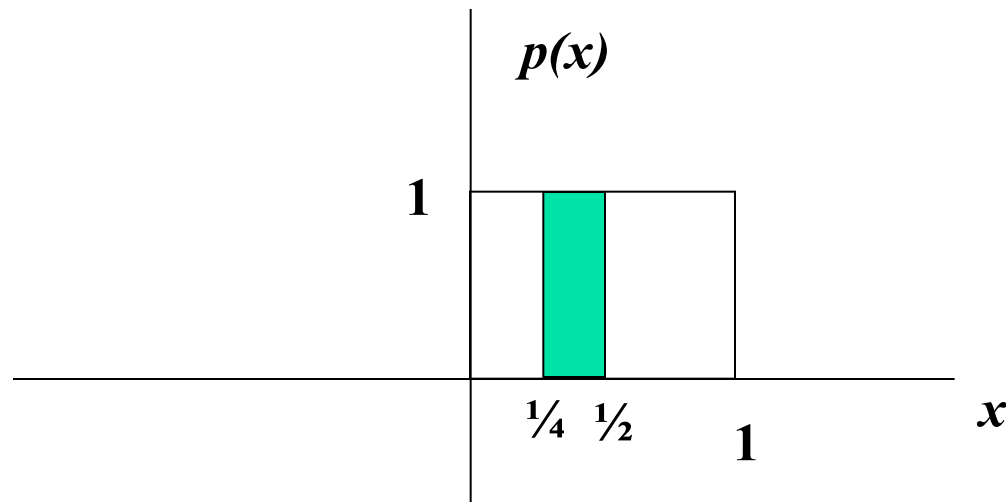


We can see it's a probability distribution because it integrates to 1 (the area under the curve is 1):

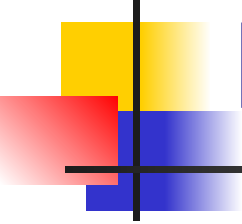
$$\int_0^1 1 = x \Big|_0^1 = 1 - 0 = 1$$

Example: Uniform distribution

What's the probability that x is between $\frac{1}{4}$ and $\frac{1}{2}$?



$$P(\frac{1}{2} \geq x \geq \frac{1}{4}) = \frac{1}{4}$$



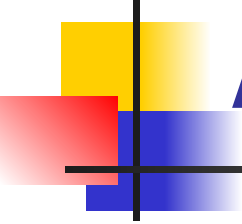
Practice Problem

4. Suppose that survival drops off rapidly in the year following diagnosis of a certain type of advanced cancer. Suppose that the length of survival (or time-to-death) is a random variable that approximately follows an exponential distribution with parameter 2 (makes it a steeper drop off):

$$\text{probability function : } p(x = T) = 2e^{-2T}$$

$$[\text{note : } \int_0^{+\infty} 2e^{-2x} = -e^{-2x} \Big|_0^{+\infty} = 0 + 1 = 1]$$

What's the probability that a person who is diagnosed with this illness survives a year?



Answer

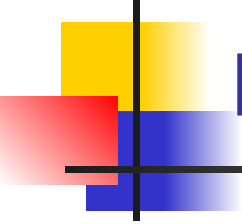
The probability of dying within 1 year can be calculated using the cumulative distribution function:

Cumulative distribution function is:

$$P(x \leq T) = -e^{-2x} \Big|_0^T = 1 - e^{-2(T)}$$

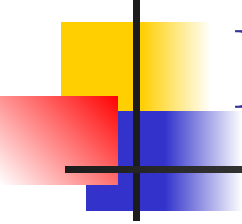
The chance of surviving past 1 year is: $P(x \geq 1) = 1 - P(x \leq 1)$

$$1 - (1 - e^{-2(1)}) = .135$$



Expected value, or mean

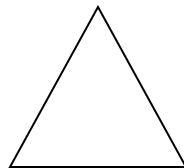
- Expected value is just the weighted average or mean (μ) of random variable x .
- Imagine placing the masses $p(x)$ at the points X on a beam; the balance point of the beam is the expected value of x .



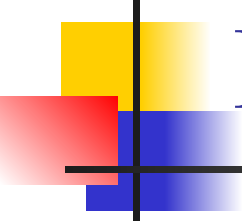
Example: expected value

- Recall the following probability distribution of ship arrivals:

x	10	11	12	13	14
$P(x)$	.4	.2	.2	.1	.1



$$\sum_{i=1}^5 x_i p(x) = 10(.4) + 11(.2) + 12(.2) + 13(.1) + 14(.1) = 11.3$$



Expected value, formally

Discrete case:

$$E(X) = \sum_{\text{all } x} x_i p(x_i)$$

Continuous case:

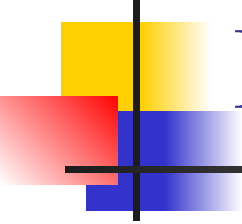
$$E(X) = \int_{\text{all } x} x_i p(x_i) dx$$

Empirical Mean is a special case of Expected Value...

Sample mean, for a sample of n subjects: =

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n} = \sum_{i=1}^n x_i \left(\frac{1}{n} \right)$$

The probability (frequency) of each person in the sample is $1/n$.



Expected value, formally

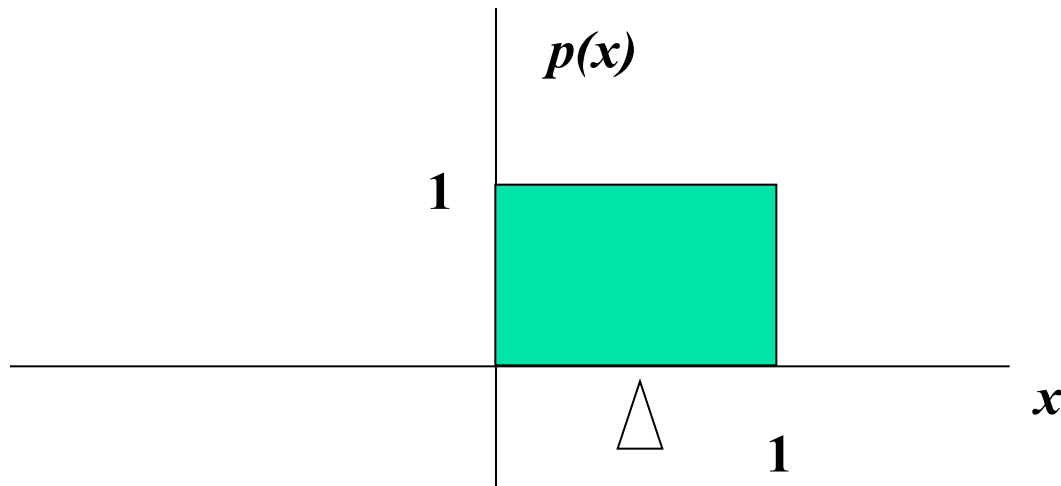
Discrete case:

$$E(X) = \sum_{\text{all } x} x_i p(x_i)$$

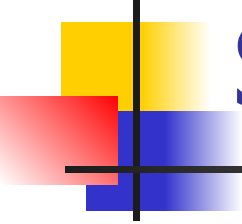
Continuous case:

$$E(X) = \int_{\text{all } x} x_i p(x_i) dx$$

Extension to continuous case: uniform distribution

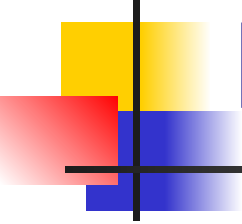


$$E(X) = \int_0^1 x(1)dx = \frac{x^2}{2} \Big|_0^1 = \frac{1}{2} - 0 = \frac{1}{2}$$



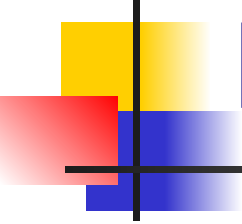
Symbol Interlude

- $E(X) = \mu$
 - these symbols are used interchangeably



Expected Value

- Expected value is an extremely useful concept for good decision-making!



Example: the lottery

- The Lottery (also known as a tax on people who are bad at math...)
- A certain lottery works by picking 6 numbers from 1 to 49. It costs \$1.00 to play the lottery, and if you win, you win \$2 million after taxes.
- *If you play the lottery once, what are your expected winnings or losses?*



Lottery

Calculate the probability of winning in 1 try:

$$\frac{1}{\binom{49}{6}} = \frac{1}{\frac{49!}{43!6!}} = \frac{1}{13,983,816} = 7.2 \times 10^{-8}$$

“49 choose 6”

Out of 49 numbers,
this is the number
of distinct
combinations of 6.

The probability function (note, sums to 1.0):

$x\\$	$p(x)$
-1	.999999928
+ 2 million	7.2×10^{-8}



Expected Value

The probability function

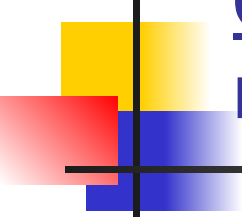
x	$p(x)$
-1	.999999928
+ 2 million	7.2×10^{-8}

Expected Value

$$\begin{aligned} E(X) &= P(\text{win}) * \$2,000,000 + P(\text{lose}) * -\$1.00 \\ &= 2.0 \times 10^6 * 7.2 \times 10^{-8} + .999999928 (-1) = .144 - .999999928 = -\$.86 \end{aligned}$$

Negative expected value is never good!

You shouldn't play if you expect to lose money!



Gambling (or how casinos can afford to give so many free drinks...)

A roulette wheel has the numbers 1 through 36, as well as 0 and 00. If you bet \$1 that an odd number comes up, you win or lose \$1 according to whether or not that event occurs. If random variable X denotes your net gain, $X=1$ with probability $18/38$ and $X=-1$ with probability $20/38$.

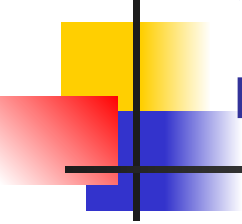
$$E(X) = 1(18/38) - 1(20/38) = -\$0.053$$

On average, the casino wins (and the player loses) 5 cents per game.

The casino rakes in even more if the stakes are higher:

$$E(X) = 10(18/38) - 10(20/38) = -\$0.53$$

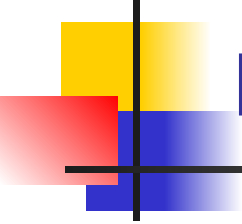
If the cost is \$10 per game, the casino wins an average of 53 cents per game. If 10,000 games are played in a night, that's a cool \$5300.



**A few notes about Expected Value as a mathematical operator:

If c = a constant number (i.e., not a variable) and X and Y are any random variables...

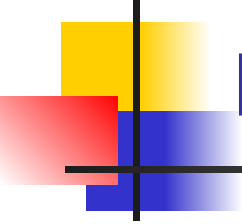
- $E(c) = c$
- $E(cX) = cE(X)$
- $E(c + X) = c + E(X)$
- $E(X + Y) = E(X) + E(Y)$


$$E(c) = c$$

$$E(c) = c$$

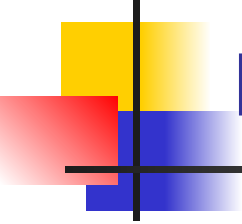
Example: If you cash in soda cans in CA, you always get 5 cents per can.

Therefore, there's no randomness. You always expect to (and do) get 5 cents.


$$E(cX) = cE(X)$$

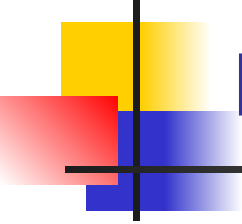
$$E(cX) = cE(X)$$

Example: If the casino charges \$10 per game instead of \$1, then the casino expects to make 10 times as much on average from the game (See roulette example above!)


$$E(c + X) = c + E(X)$$

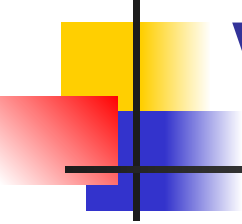
$$E(c + X) = c + E(X)$$

Example, if the casino throws in a free drink worth exactly \$5.00 every time you play a game, you always expect to (and do) gain an extra \$5.00 regardless of the outcome of the game.


$$E(X+Y) = E(X) + E(Y)$$

$$E(X+Y) = E(X) + E(Y)$$

Example: If you play the lottery twice, you expect to lose: $-\$.86$
+ $-\$.86$.

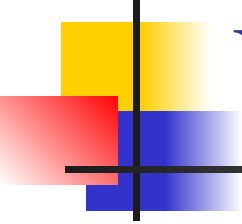


Variance/standard deviation

“The average (expected) squared distance (or deviation) from the mean”

$$\sigma^2 = Var(x) = E[(x - \mu)^2] = \sum_{\text{all } x} (x_i - \mu)^2 p(x_i)$$

***We square because squaring has better properties than absolute value. Take square root to get back linear average distance from the mean (= "standard deviation").*



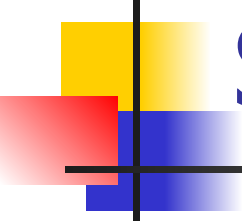
Variance, formally

Discrete case:

$$Var(X) = \sigma^2 = \sum_{\text{all } x} (x_i - \mu)^2 p(x_i)$$

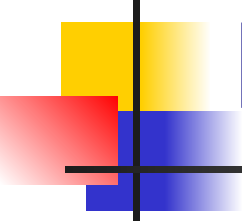
Continuous case:

$$Var(X) = \sigma^2 = \int_{-\infty}^{\infty} (x_i - \mu)^2 p(x_i) dx$$



Symbol Interlude

- $\text{Var}(X) = \sigma^2$
 - these symbols are used interchangeably



Handy calculation formula!

Handy calculation formula (if you ever need to calculate by hand!):

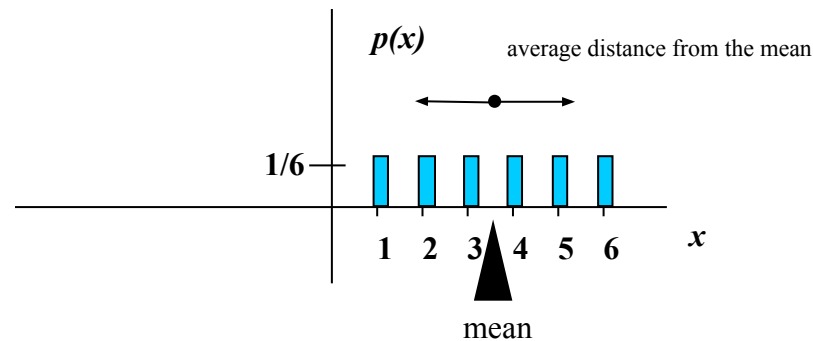
$$Var(X) = \sum_{\text{all } x} (x_i - \mu)^2 p(x_i) = \sum_{\text{all } x} x_i^2 p(x_i) - (\mu)^2$$

Intervening algebra!

$$= E(x^2) - [E(x)]^2$$

For example, what's the variance and standard deviation of the roll of a die?

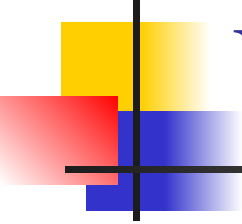
x	$p(x)$
1	$p(x=1)=1/6$
2	$p(x=2)=1/6$
3	$p(x=3)=1/6$
4	$p(x=4)=1/6$
5	$p(x=5)=1/6$
6	$p(x=6)=1/6$
1.0	



$$E(x) = \sum_{\text{all } x} x_i p(x_i) = (1)\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) = \frac{21}{6} = 3.5$$

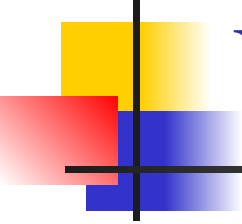
$$E(x^2) = \sum_{\text{all } x} x_i^2 p(x_i) = (1)\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 9\left(\frac{1}{6}\right) + 16\left(\frac{1}{6}\right) + 25\left(\frac{1}{6}\right) + 36\left(\frac{1}{6}\right) = 15.17$$

$$\sigma_x^2 = \text{Var}(x) = E(x^2) - [E(x)]^2 = 15.17 - 3.5^2 = 2.92$$
$$\sigma_x = \sqrt{2.92} = 1.71$$


$$\text{Var}(c) = 0$$

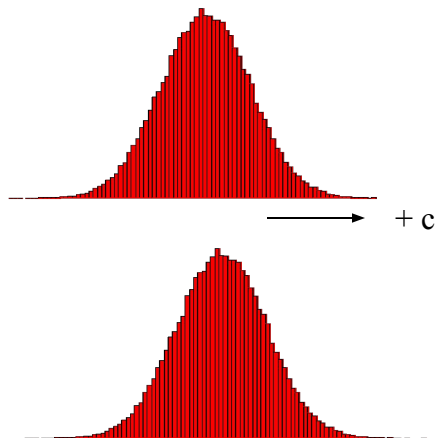
$$\text{Var}(c) = 0$$

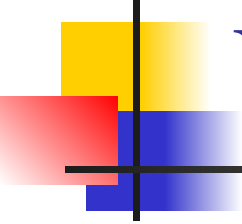
Constants don't vary!


$$\text{Var}(c+X) = \text{Var}(X)$$

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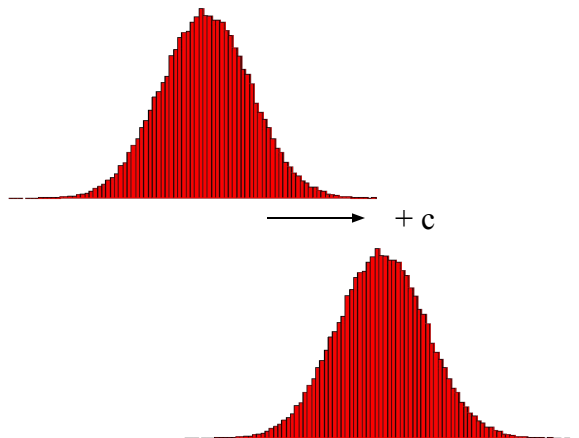
Adding a constant to every instance of a random variable doesn't change the variability. It just shifts the whole distribution by c . If everybody grew 5 inches suddenly, the variability in the population would still be the same.

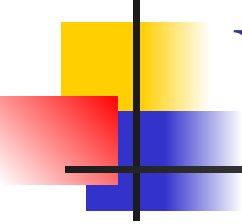



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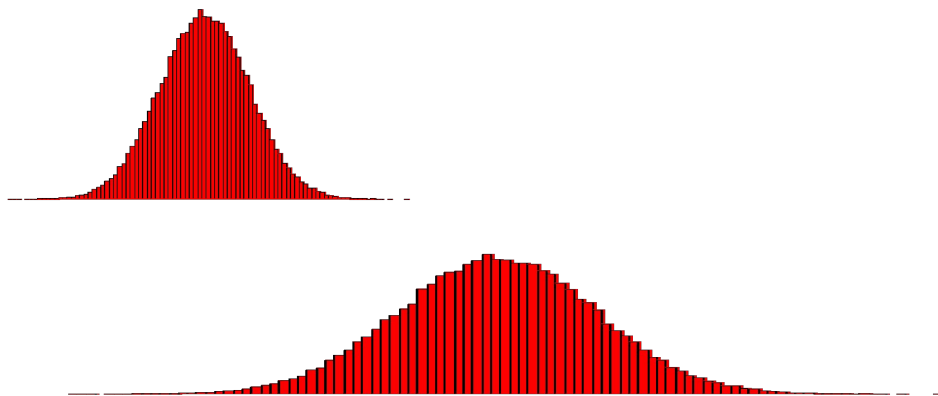
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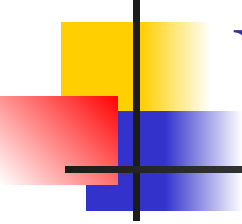



$$\text{Var}(cX) = c^2 \text{Var}(X)$$

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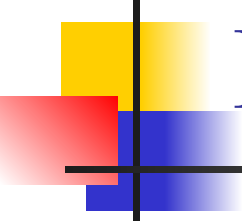
Multiplying each instance of the random variable by c makes it c -times as wide of a distribution, which corresponds to c^2 as much variance (deviation squared). For example, if everyone suddenly became twice as tall, there'd be twice the deviation and 4 times the variance in heights in the population.




$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$$

$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$ ***ONLY IF X and Y are independent!!!!!!!***

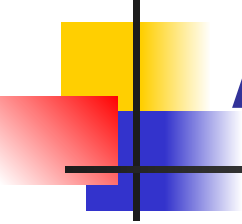
With two random variables, you have more opportunity for variation, unless they vary together (are dependent, or have covariance): $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$



Practice Problem

Find the variance and standard deviation for the number of ships to arrive at the harbor (recall that the mean is 11.3).

x	10	11	12	13	14
$P(x)$	.4	.2	.2	.1	.1



Answer: variance and std dev

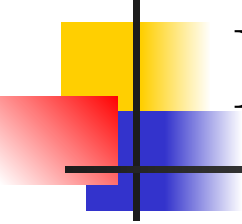
x^2	100	121	144	169	196
$P(x)$	.4	.2	.2	.1	.1

$$E(x^2) = \sum_{i=1}^5 x_i^2 p(x_i) = (100)(.4) + (121)(.2) + 144(.2) + 169(.1) + 196(.1) = 129.5$$

$$Var(x) = E(x^2) - [E(x)]^2 = 129.5 - 11.3^2 = 1.81$$

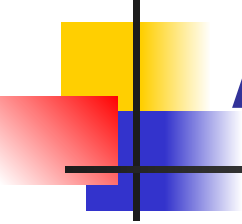
$$stddev(x) = \sqrt{1.81} = 1.35$$

Interpretation: On an average day, we expect 11.3 ships to arrive in the harbor, plus or minus 1.35. This gives you a feel for what would be considered a usual day!



Practice Problem

You toss a coin 100 times. What's the expected number of heads? What's the variance of the number of heads?



Answer: expected value

Intuitively, we'd probably all agree that we expect around 50 heads, right?

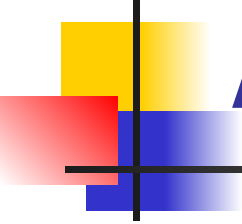
Another way to show this $\square$

Think of tossing 1 coin. $E(X=\text{number of heads}) = (1) P(\text{heads}) + (0)P(\text{tails})$

$$\therefore E(X=\text{number of heads}) = 1(.5) + 0 = .5$$

If we do this 100 times, we're looking for the sum of 100 tosses, where we assign 1 for a heads and 0 for a tails. (these are 100 "independent, identically distributed (i.i.d)" events)

$$\begin{aligned} E(X_1 + X_2 + X_3 + X_4 + X_5 + \dots + X_{100}) &= E(X_1) + E(X_2) + E(X_3) + E(X_4) + E(X_5) + \dots + \\ E(X_{100}) &= \\ 100 E(X_1) &= 50 \end{aligned}$$



Answer: variance

What's the variability, though? More tricky. But, again, we could do this for 1 coin and then use our rules of variance.

Think of tossing 1 coin.

$$E(X^2 = \text{number of heads squared}) = 1^2 P(\text{heads}) + 0^2 P(\text{tails})$$

$$\therefore E(X^2) = 1(.5) + 0 = .5$$

$$\text{Var}(X) = .5 - .5^2 = .5 - .25 = .25$$

Then, using our rule: $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$ (coin tosses are independent!)

$$\begin{aligned} \text{Var}(X_1 + X_2 + X_3 + X_4 + X_5 + \dots + X_{100}) &= \text{Var}(X_1) + \text{Var}(X_2) + \text{Var}(X_3) + \\ &\quad \text{Var}(X_4) + \text{Var}(X_5) + \dots + \text{Var}(X_{100}) = \end{aligned}$$

$$\begin{aligned} 100 \text{Var}(X_1) &= 100 (.25) = 25 \\ \text{SD}(X) &= 5 \end{aligned}$$

Interpretation: When we toss a coin 100 times, we expect to get 50 heads plus or minus 5.